In order to train our generative model *P* to generate realistic celebrity photos using data set *Q*, we need to ensure the generative model assigns high probability to the data in *Q* and also to data not in *Q* but that plausibly could be. Mathematically, we want to maximize this ratio:

$$LR = \frac{P(x)}{Q(x)}$$

We call this the *likelihood ratio* (LR) between $P(x)$ and $Q(x)$. *Likelihood* in this context is just another word for *probability*.

If we take the ratio for an example image of Will Smith that exists in *Q* using an untrained *P*, we might get

$$LR = \frac{P(x = \text{Will Smith})}{Q(x = \text{Will Smith})} = \frac{0.0001}{1.0} = 0.0001$$

This is a tiny ratio. We want to backpropagate into our generative model and do gradient descent to update its parameters so that this ratio is maximized. This likelihood ratio is the objective function we want to maximize (or minimize its negative).

But we don't want to do this just for a single image; we want the generative model to maximize the total probability of all the images in data set *Q*. We can find this total probability by taking the product of all the individual examples (because the probability of *A and B* is the probability of *A times* the probability of *B* when *A* and *B* are independent and come from the same distribution). So our new objective function is the product of the likelihood ratios for each piece of data in the data set. We have several math equations coming up but we're just using them to explain the underlying probability concepts; don't spend any time trying to remember them.

Table 7.4. The likelihood ratio in math and Python